Mzuri Organics

Training Manual

Black Soldier Fly Rearing

1. Introduction to BSF Rearing

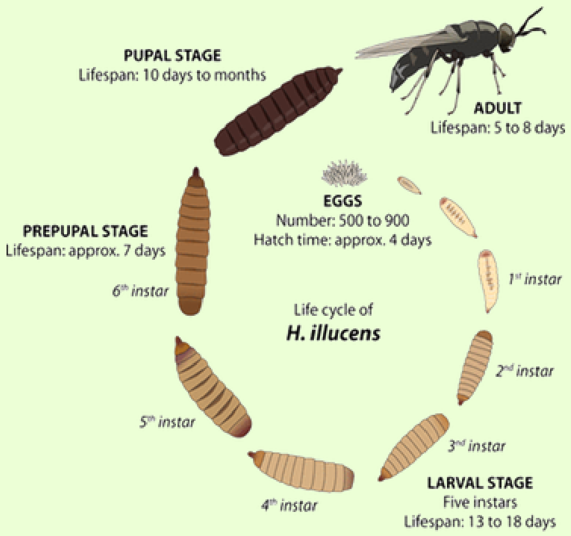
Black Soldier Fly (BSF), Hermetia illucens, is a tropical insect belonging to the Stratiomyidae family. Native to warm regions, including Kenya, BSFs are renowned for their efficiency in converting organic waste into protein-rich larvae and frass, promoting circular agriculture. Their farming offers benefits in waste management, protein production, and income generation.

2. Mzuri Organics Approach to Circularity

Our model integrates BSFL farming with vermicomposting and greenhouse cultivation. BSFL frass is processed further with red wigglers vermicompost as well as fermentation processes to produce a nutrient-dense organic biofertilizers. We also develop innovative value-added products like Green Tags, embedding tree seeds in frass-infused biodegradable paper.

3. Understanding the BSF Life Cycle

- Egg Stage: 500–1200 eggs laid near decomposing matter. Hatch in 3–5 days.
- Larval Stage: Feeds voraciously for 14–18 days. Converts waste into biomass.
- Pupal Stage: Non-feeding stage, 7–14 days. Transformation into adults.
- Adult Fly: Lives 5–14 days, mates, lays eggs, and dies.



Key Conditions:

- Temperature: 25–30°C
- Humidity: 60–80%
- Moisture: 60–70% in feed
- Light: Indirect; breeding cages need UV stimulation for mating

4. Facility Setup and Site Considerations

- Layout: Distinct zones for breeding, rearing, harvesting, and processing
- Size: 50 m^2 for breeding, 100 m^2 per ton/day of waste
- Environmental controls: Shade netting, ventilation, pest control
- Utilities: Water access, drainage, solar lighting if possible
- Safety: Buffer zones from residential areas; sanitation stations

5. BSFL Rearing Unit Operations

- Breeding: Use untreated insect net cages with oviposition traps; maintain attractants (fermented organic matter)
- Egg Incubation: Transfer eggies to moist containers; maintain humidity and label dates
- Hatching: Neonates emerge in 3–5 days; transferred to nursery bins
- Nursery Care: Start larvae on a balanced pre-feed (e.g., chick mash) until 5-DOL
- Larval Rearing: Transfer 5-DOL to feeding crates; feed on prepared substrates

6. Waste Collection and Pre-processing

Acceptable Substrates:

- Fruit/vegetable scraps, animal manure (pre-treated), brewers waste, food waste, crop residues

Preparation Steps:

- Sort organics from inorganics
- Shred/chop/grind to <2 cm or to paste
- Mix for nutrient balance (C:N ratio of 25–30:1)
- Adjust moisture (target 60–70%)
- Optional: ferment anaerobically for 48hrs

7. Larval Feeding and Growth Monitoring

- Add 200-250g 5-DOL to each larvero (60x40x17cm bin) containing 8kg substrate
- Feed schedule: Day 1, Day, Day 8-10
- Layer feed ≤8 cm deep for aeration
- Monitor temperature, humidity, and larval activity daily
- Maintain records of waste input, larval growth, and conversion

8. Larvae and Residue Harvesting

- Signs of readiness: Darkening larvae (dark brown), crawling behavior
- Manual: Use sieves to separate larvae from residue
- Self-Harvest: Set up angled trays with escape paths to collection bins
- Water Flotation: Float larvae for separation
- Weigh larvae and residue for recordkeeping

9. Post-treatment of Products

Larvae:
- Clean larvae using running water to remove residue
- Soak live larvae in clean water overnight to empty gut
- Sanitize (boiling or hot water dip)
- Dry: Solar drying, oven, or freeze-drying
- Grind: Produce BSF meal and/or oil-extract
- Pelletize: For animal feed blending
- Alternatively, after sanitizing use semi moist feed processing technology

Residue (Frass):
- Compost or vermicompost
- Mix with additives for enhanced formulations
- Use in seedling trays or field application

10. Biofertilizer and Feed Applications
- Larvae: Used in poultry, fish, pig feed; high protein/fat value (35-50%)
- BSF Oil: Alternative to soy oil in livestock feed
- Frass: Improves soil structure, microbial activity, nutrient content
- By-products: Chitin (bioplastic), peptides (pharmaceutical/cosmetic potential)

11. Hygiene, Monitoring, and Data Collection
- Clean tools, containers, and working surfaces daily
- Log sheets: Record feed amounts, temperatures, moisture, mortality, yields
- Use PPE at all stages: boots, gloves, lab coats
- Sanitize zones between operations
- Manage pests: use fly traps and fumigate as needed

12. Worker Safety and Protective Measures

- PPE: Gloves, boots, masks, overalls/dust coats
- Handwashing stations at exit points
- Waste segregation and safe handling protocols
- Routine disinfection and monthly pest control
- Training on zoonotic disease prevention and first aid

13. Appendices

Appendix A: Standard Operating Procedures (SOPs)
Appendix B: Daily Checklist Templates
Appendix C: Pictures

APPENDIX A: STANDARD OPERATING PROCEDURES

Egg Collection and Incubation

Purpose: Ensure a continuous and healthy supply of larvae by collecting and incubating BSF eggs under optimal conditions.
Scope: Applies to breeding and hatchery staff.
Materials and Equipment: Egg collection traps (eggies), hatchling containers, humidifiers/watering containers, thermo-hygrometers, labels, and recording sheets.
Procedure: Collect eggies every-other-day from oviposition sites, label with date, and transfer to a humid hatchling area. Maintain relative humidity between 60-70%. Check hatching progress after 3-5 days.
Monitoring and Records: Daily humidity checks; hatch rate records.
Health and Safety Notes: Wear gloves; sanitize equipment weekly.

Larval Rearing Management

Purpose: Maximize larval growth and conversion efficiency.
Scope: Applies to nursery and rearing teams.
Materials and Equipment: Larveros, feed substrates, thermometers, weighing scales.
Procedure: Transfer neonates into larveros. Feed on Days 1 and 8. Maintain substrate moisture at 60–70% and depth under 8cm. Monitor larval activity and temperatures daily.
Monitoring and Records: Waste intake, growth measurements, mortality rates.
Health and Safety Notes: Wash hands before and after handling larvae.

Waste Collection and Preprocessing

Purpose: Prepare safe, nutrient-balanced substrates for larval feeding.
Scope: Applies to waste processing staff.
Materials and Equipment: Shredder, sorting tables, buckets.
Procedure: Sort incoming waste to remove plastics and metals. Shred organic matter to <2cm particle size. Blend multiple sources for nutrient balance. Adjust moisture to 70% using water if needed.
Monitoring and Records: Record substrate types and quantities daily.
Health and Safety Notes: Use gloves, boots, and masks when handling waste.

Larvae Harvesting and Post-Treatment

Purpose: Harvest BSF larvae efficiently and prepare them for processing or sale.
Scope: Applies to rearing and processing staff.
Materials and Equipment: Sieves, collection bins, drying trays.
Procedure: Identify dark-colored (dark brown) larvae. Harvest by manual sieving. If turned to pre-pupae self-harvest system can be more efficient.
Clean and sanitize larvae.
Dry larvae to <10% moisture by solar or oven drying.
Monitoring and Records: Harvest weights; drying temperatures.
Health and Safety Notes: Handle hot drying trays with protective gloves.

Residue and Frass Handling

Purpose: Manage frass to produce safe, nutrient-rich biofertilizer.
Scope: Applies to frass management teams.
Materials and Equipment: Composting bins, moisture meters.
Procedure: Transfer residue after harvesting to composting units. Turn frass piles every 3 days. Monitor moisture and temperature during composting.
Monitoring and Records: Compost turning schedule; temperature logs.
Health and Safety Notes: Wash hands after handling frass.

Facility Hygiene and Sanitation

Purpose: Maintain a clean and safe working environment.
Scope: Applies to all staff.
Materials and Equipment: Brushes, detergents, disinfectants.
Procedure: Daily sweeping and cleaning of all rearing and processing areas. Weekly deep cleaning of empty cages, larveros, and working surfaces.
Monitoring and Records: Cleaning log sheets.
Health and Safety Notes: Use proper PPE during cleaning activities.

Worker Safety and PPE Usage

Purpose: Protect workers from injuries and health hazards.
Scope: Applies to all personnel.
Materials and Equipment: PPE (boots, gloves, goggles, dust coats).
Procedure: Ensure PPE is worn at all times when handling waste, larvae, or frass. Replace torn or damaged PPE immediately.
Monitoring and Records: Monthly PPE inspection reports.
Health and Safety Notes: First-aid kits must be readily available.

Pest Control in BSF Rearing Facilities

Purpose: Prevent infestation of pests that can harm BSF colonies.
Scope: Applies to facility maintenance teams.
Materials and Equipment: Fly and rodent traps, sealing materials.
Procedure: Inspect facility weekly for pests. Install traps near breeding areas.
Monitoring and Records: Pest inspection logs.
Health and Safety Notes: Follow safe fumigation practices; evacuate area during treatments.

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APPENDIX B: CHECKLISTS

1.Daily Operations Checklist

- Collect and label eggies
- Hatchling transfer and humidity check
- Feed larvae (Day 1/8 -10 as scheduled)
- Sort and shred incoming waste
- Adjust substrate moisture
- Harvest prepupae if ready
- Record waste intake and larval growth
- Clean equipment and workspaces
- Pest control inspections
- PPE usage check

APPENDIX B: CHECKLISTS

2. Weekly Facility Inspection Checklist

- Inspect breeding cages for integrity
- Inspect larveros for damage
- Clean love cages and rearing trays
- Check pest traps and take necessary action
- Review hygiene logs
- Conduct PPE inventory

Weekly Production Summary Template

Date: ..../.............../..................

Total larvae produced: ___________ kg
Total frass generated: ___________ kg
Mortality rates: ___________ %
Product sales: ___________ KES

Supervisor Name: .... Sign: ....

APPENDIX B: CHECKLISTS

3. Weekly Production Summary Template

Date: ..../.............../..................

Total larvae produced: ___________ kg
Total frass generated: ___________ kg
Mortality rates: ___________ %
Product sales: ___________ KES

Supervisor Name: .... Sign: ....

APPENDIX B: CHECKLISTS

4. Monthly Production Summary Template

Date: ..../.............../..................

• Total larvae produced: ___________ kg
• Total frass generated: ___________ kg
• Mortality rates: ___________ %
• Product sales: ___________ KES
• PPE condition report submitted: [Yes/No]
• Equipment maintenance conducted: [Yes/No]

Supervisor Name: .... Sign: ....

APPENDIX C: PICTURES

VermiFrass Compost

Integrated bsfl and crop farming

Frass and larvae separation/sieving

Egg collection devices/eggies

Waste processing/crushing

Worker safety and health/PPE

Product marketing/exhibition

Integration of bsfl and vermicomposting

Product packaging

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PRODUCTS

Frass green tags

Coriander planted
with bsfl fertilizer.

Tomatoes and kales
planted with
VermiFrass

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Dried bsfl

Bsfl frass

Organic waste

Building collaborations

Community Engagement

Investor engagement

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Larvarium

Love cages

Mzuri®
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Bsfl extension service

bsfl fertilizer testing

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Live bsfl

100% prepupae

70% prepupae

THE

End

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